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Cultivating resilience - The case for depth before scaling in agricultural insurance

1 message

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THE Behavioral Lens



Fairness bias

People strongly prefer fairness and will act to correct unfairness even at a personal cost.



03/26

Cultivating resilience - The case for depth before scaling in agricultural insurance

By Rahab Kariuki,

I have never met anyone who doesn't want insurance right after a loss. For instance, when severe floods struck parts of Nairobi in April 2024, nearly everyone wished they had insurance to help them recover. Conversations on social media reflected a widespread sentiment: many people would have eagerly bought coverage if they could turn back time. However, statistics show that less than 3% of Kenyans hold any form of insurance, even when it's both available and affordable. This gap between what people desire after disaster and the choices they make beforehand highlights the persistent challenge of why insurance uptake is so low where it's needed most?

Yet, despite the potential of climate insurance to serve as a crucial safety net for smallholder farmers in Africa since its introduction two decades ago, many national agriculture insurance programs remain in their early stages and struggle to achieve effective scale. This paradox is worth exploring, as it highlights the disconnect between the desire for insurance and its actual uptake. With over 15 years of experience in agricultural index insurance and behavioral science, I have witnessed both the rise and fall of various index insurance programs. Given the increasing impact of climate change on agricultural productivity, it is urgent that we make index insurance work for the majority of farmers and stakeholders to enhance resilience and secure livelihoods across the continent.

Few African countries, including Zambia, Uganda, Kenya, Rwanda, and Senegal, have

successfully established national crop insurance programs. Meanwhile, others, such as Ethiopia, Tanzania, Mali, Mozambique, and Benin, are still in the pilot stages of development. A notable example of a program that has achieved scale is Zambia's Farmer Input Support Programme (FISP), which has insured 1,024,434 farmers annually since 2017 according to the 2023/2024 implementation handbook. The success of Zambia's FISP program shows that large-scale insurance coverage is possible when integrated into a broader government agricultural support package. It also highlights the importance of strong public-private partnerships and effective delivery systems in reaching millions of smallholder farmers. Although these programs have demonstrated some success, there remains significant potential for growth. For example, Kenya's program has struggled to find a sustainable funding model. Farmer subsidies are available in some places, but in others, the program remains unfunded.

Looking at these case studies, literature identifies three primary barriers contributing to low uptake—basis risk, high costs, and low literacy (Ali et al., 2020; Carter, 2017; Smith, 2019)—it is important to note that the extent of government support varies considerably among these countries which in turn contributes to additional bottlenecks in scaling. Common recommendations to address these issues, including improved design, enhanced education, and increased government support (Nshakira-Rukundo et al., 2021; Greatrex et al., 2015), often overlook the critical role of human behavior in the uptake of these already complex products.

I align with those who advocate for tailoring insurance solutions to local contexts and farmer preferences (Jensen and Barrett, 2017), as well as combining insurance with disaster assistance and safety nets to broaden protection (Kramer et al., 2022; Greatrex et al., 2015).

The narrative surrounding these challenges is often oversimplified. For instance, financial literacy is frequently cited as a barrier; however, behavioral science reveals that merely providing information does not necessarily lead to a change in the way people operate. This is largely due to the classic intention-action gap, which persists when we focus solely on delivering information without addressing the underlying motivations and emotional responses that drive decision-making. Interestingly, offering background information on how an index is derived or payouts computed will only increase resistance to purchase (Elabed and Carter, 2015).

However, I argue that the crucial question isn't why these programs aren't scaling, but rather whether there is substantial evidence of impact that justifies scaling.

Factors such as risk aversion and value perceptions pose a more significant influence on individuals' decisions than the mechanics of the product itself (Brick et al., 2015). Additionally, many programs fail to engage adequately and consistently with farmers, neglecting the importance of understanding their unique contexts and needs. By not involving farmers in the co-design process, these programs miss the opportunity to address the psychological and behavioral dimensions that truly impact uptake.

Therefore, a more nuanced approach that incorporates behavioral insights alongside structural considerations is essential for creating effective and sustainable insurance solutions. This article presents an alternative perspective on scaling these initiatives based on my experience working in agriculture insurance and now, applied behavior science research in insurance.

From my analysis of successful case studies like India and Zambia, I have identified a three-

stage maturation model for index insurance programs:

Stage	Description	Key thresholds to advance
Foundational policy initiatives stage	Establishes the essential regulatory frameworks and institutional support for index insurance programs.	- Initial pilot success: Successful implementation of pilot projects showing feasibility. - Regulatory frameworks: Development of laws and regulations supporting index insurance, ensuring legal backing and consumer protection.
Data-driven innovation and efficiency stage	Focuses on leveraging data analytics and technology to improve insurance product effectiveness.	- Increased demand from stakeholders: Clear demand from farmers and agricultural organizations for tailored products. - Investment in technology: Adoption of advanced data collection and analysis tools for better product design and delivery.
Adaptive learning and continuous improvement stage	Embraces dynamic feedback loops for ongoing refinements and scaling of insurance programs.	- Demonstrated impact: Evidence of improved farmer resilience and satisfaction such as higher crop yields, financial stability. - Increased trust among farmers: Established reputation for reliability and effectiveness, leading to greater participation and investment.

In my assessment most African schemes appear to be stuck in the first two stages, which hinders their growth potential. To advance these programs to stage three, I propose a focus on ‘increasing depth before scaling’. Here, I outline three key arguments on how to achieve this depth in the ‘adaptive learning and continuous improvement’ stage.

Insured smallholder farmers are severely underinsured

A critical gap exists in the ratio of insurance coverage to the actual risk exposure faced by farmers. Many existing schemes adopt a bundling approach that primarily focuses on insuring the cost of inputs—such as seeds, fertilizers, and input loans—rather than addressing the true

exposure, which is the value of the expected yield. Farmers are fundamentally in the business of producing a harvest and aim to earn more than just breakeven.

Unfortunately, these bundling strategies and partnerships often fail to track individual farmers' data, making it challenging to enhance retention and re-enrollment rates among insured farmers. Even in the successful case of the Zambia FISP program, if the government subsidy were to be withdrawn, would the over one million insured farmers renew their policies? If insurance merely protects banks and input providers—who, by the way, do not purchase agricultural insurance for their own exposure—how does it genuinely protect and facilitate farmers' businesses and livelihoods?

The emphasis on bundling insurance products with input loans often results in a narrow focus that overlooks the broader needs of farmers. While bundling may seem like a straightforward approach to scaling, it can inadvertently limit the scope of coverage and fail to address the comprehensive risks farmers encounter. By aligning insurance products with the real risks farmers face, we can build trust and encourage wider participation in these programs.

Insurance product communication is confusing

Based on the above point, if we are not falling back on simple bundling partnerships, a crucial factor in achieving depth is the effective communication of product offerings. Farmers often receive marketing information under the guise of financial literacy, but these training sessions typically last no more than an hour and are often delivered by unqualified and untrusted representatives. This situation leads to increased ambiguity, heightened risk aversion, confusion and reluctance to adopt new products. Given that as an industry we rarely develop new insurance products, it is crucial to recognize that the primary gap in insurance lies in communication. Investing in message testing, well designed communication campaigns that simplify and demonstrate how these products meet farmers' needs is vital.

The real barrier lies not in financial literacy but in demonstrating the value of index insurance. Much like how individuals use mobile money services without fully understanding their mechanics, farmers can engage with climate insurance if they perceive its benefits clearly. The challenge is conveying how these products can mitigate risks and enhance financial stability. When mobile money services were introduced in Kenya, early users tested the system by sending small amounts and confirming receipt before sending larger sums. This gradual approach allowed them to build confidence in the system. For insurance, however, the testing process takes longer, and an underinsurance mistake can be costly if a payout is needed. To mitigate these risks, it's crucial to provide clear, practical demonstrations of how index insurance works and how it can deliver tangible benefits to farmers' lives.

To do this, insurers should focus on relatable examples and practical demonstrations that resonate with farmers' experiences. For instance, simply reframing livestock index insurance as a policy designed to cover household expenses during a drought reveals significant differences in demand among women in Northern Kenya (Hobbs, 2019). Furthermore, the use of interactive tools, such as AB Entheo's ResilientMe! and UC Davis SimPastoralist games, has proven to be a more effective approach.

Moreover, leveraging trusted individuals—such as local leaders or fellow farmers who have benefited from index insurance—as advocates can enhance credibility. Social proof through

community testimonials can significantly increase engagement, making farmers more likely to participate in these programs.

Insurance outreach is limited primarily to smallholder producers

Lastly, we need to consider expanding the application of index insurance beyond smallholder producers and beyond agriculture. Currently, many schemes focus solely on encouraging individual farmers to insure their production neglecting County governments, development partners, input providers and offtakers among other stakeholder which limits the business case and sustainability of agriculture insurance. By insuring all value chain actors—suppliers, distributors, and processors—we can spread the risk and make the cost of insurance more manageable for farmers.

Climate risks affect many industries including healthcare, energy, transportation, finance, real estate, tourism, manufacturing, construction, and technology among others. Index insurance can be applied as an innovative financing and safety net mechanism in these areas too. This broader approach not only enhances the sustainability of insurance programs but also promotes a more resilient agricultural ecosystem. For instance, integrating health insurance solutions into agricultural support efforts provides a holistic approach to farmer welfare. By protecting not only the farmers but also their households and entire value chains, we can create a more stable environment that reduces risk for everyone involved.

Lessons from microfinance and national health insurance

The concept of increasing depth before scaling has been notably successful in the field of microfinance, particularly in developing countries. For instance, organizations like Grameen Bank initially focused on understanding the financial behaviors and requirements of low-income individuals, which allowed them to design products that were not only accessible but also build trust and engagement. This deep engagement ensured that when they scaled their operations, they did so with a solid foundation of client understanding and product relevance.

Similarly, national health insurance programs have demonstrated the importance of robust government support and well-defined policy frameworks. By showing clear and immediate benefits, these programs have built strong public trust. Agricultural insurance schemes can learn valuable lessons from these successful models to overcome common barriers. For example, just as health insurance programs utilize subsidies from private sector and government levies to encourage enrollment among low-income populations, agricultural insurance can implement sustainable premium support measures to alleviate the financial burden on smallholder farmers.

As I conclude, I cannot help but think back on my journey in the agricultural index insurance sector. I am reminded of a smallholder farmer named Jane, whom I met during my first week of fieldwork with Kilimo Salama in Laikipia, Kenya. Jane had dedicated her life to cultivating her family's land, carefully nurturing her crops and saving every penny she could. When she first heard about index insurance, she was hopeful it would provide the safety net she desperately needed. However, like many farmers, she found the seed insurance payouts inadequate, found the products confusing, and felt they did not sufficiently cover her actual risks.

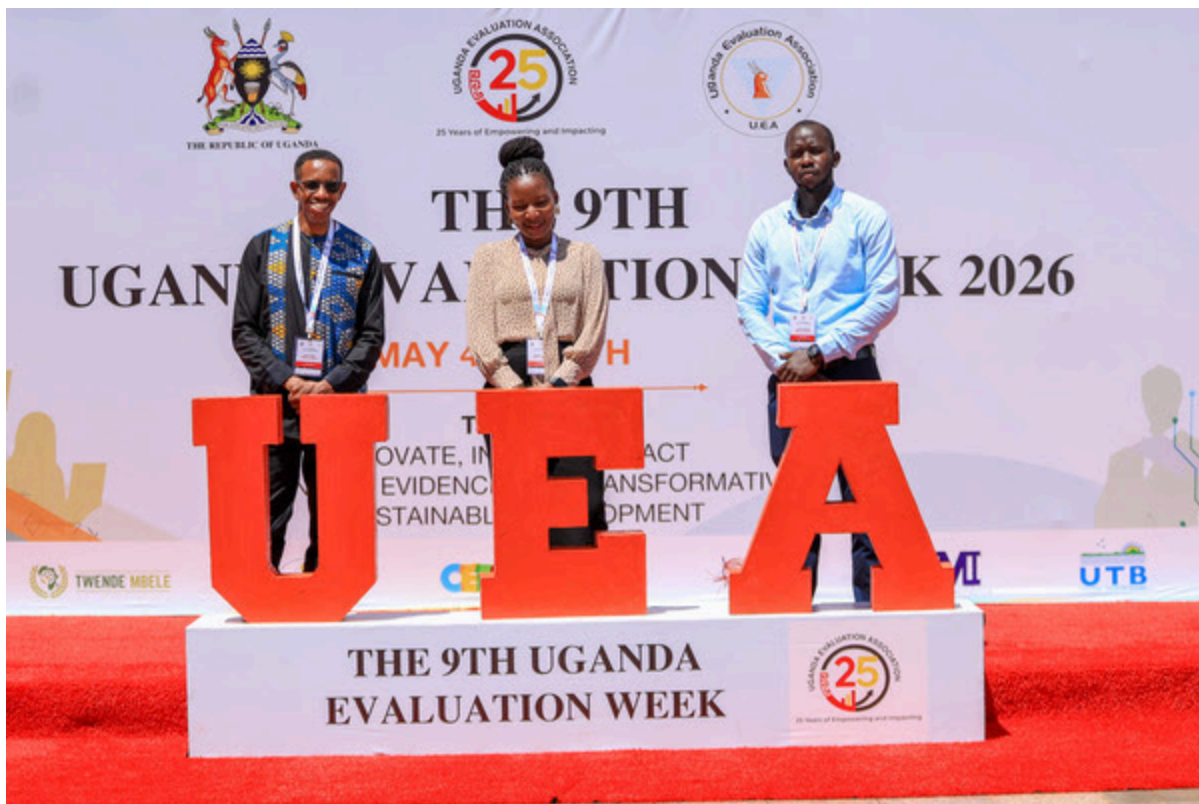
While this encounter was 16 years ago, Jane's experience illustrates the urgent need for change in our approach to climate insurance. We must ensure that farmers like her can access coverage

that truly reflects their vulnerabilities. By focusing on increasing depth through tailored products, simplified communication, and broader applications, we can empower farmers to protect their livelihoods effectively.

This is not just about scaling insurance programs; it's about creating meaningful solutions that resonate with the realities of farmers' lives. We have the opportunity to make a lasting impact, to transform index insurance from a mere concept into a powerful tool for resilience.

Insurance is not just a product, but a lifeline. By prioritizing depth before scaling, we can ensure that every farmer has the opportunity to thrive, just like Jane.

News and events



The Busara team participated in the Uganda Evaluation Week 2026.



The Busara delegate at the On Think Tanks Conference 2026 in Rabat, visited the Kenyan Embassy in Morocco.



Busara's Rowan Harrity in the company of other representatives from Sudan, Haiti, Kenya, Niger, Somalia, South Sudan, Sudan, Syria, Yemen and the Central African Republic, as well as technical leadership from UNICEF, Gavi, Gates Foundation, WHO and Common Thread joined for a Global Vaccination Demand meeting.



Busara was fully represented at the STRI4Society Week 2026, with Nimo Muthike contributing to

a breakout session titled “Strengthening Research and Policy Linkage in Kenya’s Research Ecosystem.”

From the B.R.A.C.E team

Coins and honesty: A look at the coin flip (or dice) honesty task

Author: Raphael Malenya

The coin flip honesty task is a behavioral economics experiment that measures honesty by allowing participants to privately report coin flip outcomes tied to rewards. Because researchers cannot observe the actual result, participants have an opportunity to lie for personal gain. Researchers have adapted the task through repeated trials, public versus private reporting, and group settings to study different influences on honest behavior.

A study across 15 countries found that honesty levels varied significantly between populations and were linked to historical economic development. It also showed that people are often poor judges of how honest individuals from other countries are, with perceptions shaped by cognitive biases.

The task is popular because it is simple, flexible, and easy for participants to understand. However, factors such as cultural norms, reward size, social desirability bias, and the gap between experimental settings and real-life ethical decisions can affect results.

Read more [here](#)



On the web

[When germs start adapting: Reflections from the 2026 Antimicrobial Resistance Summit in Sydney, Australia](#)

There's a joke that keeps showing on my feed. It centers around this idea that if an average person actually went back in time, they would be very incapable of proving they are from the future. While they will be able to speak of technologies such as computers and phones, they would neither be able to replicate, nor explain how they work. Most of us would face the same issue, having come up in a highly globalized, industrialized world with increasing distance between specializations, we rely on each other without even knowing it. Nothing worth doing ends up getting done by a singular actor.

Read more [here](#)

[A new operating system for philanthropy. Reflections from the Global Philanthropy Forum 2026](#)

Many of the world's biggest problems, such as poverty, climate change, education, health, and governance failures, remain stubborn despite decades of philanthropic funding. Philanthrists are increasingly realizing that pouring more money into individual projects is not necessarily producing lasting change because the underlying systems, incentives, and behaviors that create these problems remain intact. As challenges become more interconnected and complex, there is growing pressure on the philanthropy sector to move beyond traditional grantmaking and find better ways to create systemic impact.

Read [more](#)

[Thumbs up or down: Regulating social media is a double-edged sword](#)

Social media has become one of the most powerful tools for information sharing, political mobilization, and citizen engagement across Africa. It has helped citizens organize movements, hold leaders accountable, and access information, but it has also enabled misinformation, manipulation, polarization, and political influence campaigns. As governments struggle to respond, many are introducing regulations that risk restricting free expression and civic participation. The challenge is finding a balance between protecting citizens from online harms and preserving the democratic benefits that social media has created.

Read [more](#)

[Verifying love in the lab: lessons from a financial privacy study](#)

Financial privacy within relationships is often misunderstood. While people commonly assume that partners who hide money, savings, or spending from each other are acting out of distrust or deception, the study found that financial privacy can sometimes be a way for individuals, especially women, to maintain autonomy, manage competing household priorities, or protect themselves from financial pressures. This raises a broader question: how much financial transparency is healthy in a relationship, and when does privacy strengthen rather than undermine trust?

Read more [here](#)

[The slfA tool for a Statement of Intellectual Fellowship and Accountability](#)

The release of the [slfA \(Statement of Intellectual Fellowship and Accountability\) tool](#) comes at a time when AI is becoming deeply embedded in research and knowledge production, but norms for transparently disclosing how AI is used remain underdeveloped. The tool was created to help researchers reflect on and document human-AI interactions across different stages of a project, making contributions, accountability, and AI usage more visible. By combining AI disclosure with the CRediT contributor framework, slfA aims to foster a culture of transparency, learning, and responsible knowledge production as AI becomes a routine part of research workflows.

Read more [here](#)

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“Man is a social animal... He desires not only to be loved, but to be lovely.”

- Adam Smith



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